

Energy resources

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 13 starred (★) questions first.

	QUESTION	ANSWER	✓
1	Name the three main uses for energy resources.	Transport, electricity generation and heating	☐
2 ★	What is a renewable energy resource?	One that is being (or can be) replenished as it is used.	☐
3	Name the nine main energy resources.	Fossil fuels, nuclear fuel, biofuel, wind, hydroelectricity, geothermal, tides, the Sun and water waves	☐
4 ★	Name the three types of fossil fuels.	Coal, oil and natural gas	☐
5 ★	Identify the two main non-renewable energy resources.	Fossil fuels and nuclear fuel	☐
6	Name two energy resources which involve burning or combustion.	Fossil fuels and biofuel	☐
7 ★	State what reliable means for an energy resource.	It can supply energy whenever it is needed.	☐
8 ★	Explain why solar power is not reliable in every country.	The amount of sunlight varies with the weather and the time of day.	☐
9	Explain why wind power is not reliable in every country.	The wind speed varies, and sometimes there is no wind.	☐
10 ★	Explain why fossil fuels are considered reliable.	They can be burned at any time, in varying amounts, to match demand.	☐
11	Explain why hydroelectricity is considered reliable.	The flow of water through a dam can be adjusted depending on demand.	☐
12	Identify the energy resource that contributes most to climate change.	Fossil fuels	☐

	QUESTION	ANSWER	✓
13	Other than climate change, name two environmental problems that energy resources can cause.	Any two from: habitat destruction, air pollution, radioactive waste, visual or noise pollution	☐
14	Name two energy resources that can fuel internal combustion engines.	Fossil fuels and biofuel	☐
15	Name the energy resources that can be used for electricity generation.	Fossil fuels, nuclear fuel, biofuel, wind, hydroelectricity, solar, geothermal, tidal and wave	☐
16	Name the energy resources that can be used for heating.	Fossil fuels, biofuel, geothermal and solar	☐
17	State four types of consideration that affect the choice of energy resource.	Environmental, political, social and economic	☐
18 ★	What is a non-renewable energy resource?	One that is not replenished as it is used, so it will run out.	☐
19	Explain why nuclear fuel is a non-renewable energy resource.	The supply of uranium and plutonium is finite and is not replenished.	☐
20 ★	Give one advantage of nuclear power over burning fossil fuels.	It produces no carbon dioxide when generating electricity.	☐
21 ★	Give one disadvantage of nuclear power.	It produces radioactive waste that stays hazardous for a very long time.	☐
22 ★	Name the gas from burning fossil fuels that contributes most to climate change.	Carbon dioxide	☐
23 ★	Other than climate change, state one environmental problem caused by burning fossil fuels.	Acid rain (caused by sulfur dioxide)	☐
24	State one environmental impact of a hydroelectric dam or tidal barrage.	It floods land or changes habitats, harming wildlife.	☐
25 ★	Explain why tidal power is more reliable than wind power.	The tides are predictable; the wind is not.	☐
26	Explain why scientists cannot always deal with the environmental issues they identify.	Because of political, social, ethical or economic considerations.	☐

QUESTION		ANSWER	✓
27 ★	Explain why biofuel can be described as carbon neutral.	The plants absorb as much carbon dioxide growing as is released when the fuel burns.	☐
28	Give one disadvantage of geothermal power.	It is only possible in a few places.	☐
29	Compare the costs of renewable energy resources with fossil fuels.	Expensive to set up, but no fuel cost and cheap to run.	☐